Practice Materials: Data Science Fundamentals

Exercises and self-check | Sample material

Exercise 1: Data types

Classify the following as numerical, categorical, text, or time-series data: student age, branch, review comment, daily temperature.

Exercise 2: Data cleaning

A CSV contains blank marks, repeated student IDs, and dates in two formats. List a safe sequence of cleaning steps.

Exercise 3: Visualization

Choose a chart to compare marks across subjects and explain why it is appropriate.

Exercise 4: Model selection

Give one example each of a classification problem and a regression problem.

Self-check

- Did you explain your choices?
- Did you check assumptions?
- Did you note limitations?
- Can another student reproduce your steps?

Sample educational material for the MBU DS LAB Resources page. Verify course-specific requirements with your faculty.